

2.3 Sequences *continued*

Subject content	Guidance	Curriculum reference
D Generate terms of a sequence using a position-to-term rule in context	The position-to-term rule of a sequence is: 'Double the position number then subtract 1' Find the first 5 terms of this sequence.	A7.2D
E Use the n th term to generate a linear or quadratic sequence	Find the first 5 terms of the sequences: $5n + 7$ $4n^2$	A7.2E A9.4A
F Find the n th term of an arithmetic sequence	Find the n th term of the sequence: 2, 9, 16, 23, 30, ...	N9.4B
G Recognise and continue more complex geometric sequences	Find the next term in the sequence: 1, 2.5, 5, 8.5, 13, ...	A9.4C
H Solve problems involving sequences	Ali writes an arithmetic sequence that starts with: 4, 7, 10, 13, ... Bhavin writes an arithmetic sequence that starts with: 15, 23, 31, 39, ... Ali says that 111 is in both of their sequences. Is he correct? You must show your working.	A7.2F A9.4D

2.4 Graphs

Subject content	Guidance	Curriculum reference												
A Recognise, name and plot graphs parallel to the axes and the graphs of $y = x$ and $y = -x$	Plot the graphs: $x = 4$ $y = -3$	A7.3A												
B Plot linear graphs using a table of values	Plot the graph $y = 2x + 3$ <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td></td><td></td><td>7</td><td></td><td></td></tr></table>	x	0	1	2	3	4	y			7			A7.3B
x	0	1	2	3	4									
y			7											
C Find the midpoint of a line segment, given two coordinates	Find the midpoint of a line with ends at $(-2, 3)$ and $(6, 10)$	A7.3D												
D Solve problems involving coordinates and straight lines	Given three coordinates A, B, C , write down the coordinates of D such that $ABCD$ is a parallelogram.	A7.3E												
E Recognise, draw and interpret graphs showing constant rates of change	A tap runs at a constant rate. After 8 minutes, the tap has run 75 litres of water. Plot this information on a graph. Use your graph to estimate how long it took to run 40 litres of water.	A7.3C A8.3A A9.5A												
F Draw, use and/or interpret distance-time graphs	Vladimir leaves home at 08 00 He travels for 30 minutes at a constant speed until he reaches a shop, 3 km away. He stays at the shop for 45 minutes, then returns home arriving at 09 35 Plot this information on a distance-time graph. Which part of his journey was the fastest? Explain how you know this from your graph.	A8.3B												
G Solve problems by drawing and interpret linear conversion graphs	Currency conversion graphs	A8.3A A8.3C												
H Solve problems by drawing and interpreting real-life non-linear graphs	Read values and describe trends from a real-life graph	A7.3C A8.3D												